Interaction terms improve tabular classification on synthetic benchmarks

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2022

Abstract

On a synthetic two-class set whose label depends on a linear part and one pairwise interaction between two features, a model that also sees the interaction term reaches a higher held-out AUROC than one that sees only the linear features. This paper is one of two written for a test fixture, never a paper anyone submitted anywhere.

1 Method

Two features are drawn uniformly and multiplied; the product is added to the model's inputs alongside the linear features already there.

2 Result

The interaction term raises held-out AUROC by several points over the linear-only model, on every signal strength tried.